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M 301 HEAT RECOVERY STEAM GENERATOR AND

AUXILIARY EQUIPMENT

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M 301 HEAT RECOVERY STEAM GENERATOR AND AUXILIARY EQUIMENTS

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with Part 4, subsection 1.2 “Summary of Work”.

- For each Combined Cycle Block; the Bidder shall supply two (2) Heat Recovery Steam Generator (HRSG) units for block 2 and three (3) Heat Recovery Steam Generator (HRSG) units for block 3 and three (3) Heat Recovery Steam Generator (HRSG) units for block 4. HRSG units should be of natural or forced circulation, reheat, two or three pressure level drum type and vertical or horizontal used in Combined Cycle Gas and Steam Turbine Power Plant.
- Each Combined Cycle Block shall comprises of two (2) Heat Recovery Steam Generators assigned to one (1) Steam Turbine for block 2 and three (3) Heat Recovery Steam Generators assigned to one (1) Steam Turbine for block 3 and three (3) Heat Recovery Steam Generators assigned to one (1) Steam Turbine for block 4

The HRSG should be completed with associated items but not limited to the following: Economizer, Evaporator, Drum, Superheater, Ducting, Stack, Structural and Steel works, Piping, Valves and Support, Damper and Expansion joints, Thermal insulation and Lagging, HRSG water, Circulation system including pumps (if required), Vents, Drains and Blowdown system, Platforms, Walkway stairs, Ladder handrails and all other appurtenances for a complete and operable unit as part of the Combined Cycle Power Plant.

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2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4 Subsection 2.5 “Codes and Standards”

The up-to-date Codes and Standards of the following organizations extracted from Part 4 Section 2.5 “Codes and Standards” are specifically applicable to the design, manufacture and testing of the equipment included in this specifications. Not only American Standards but also equivalent code and standards are applicable.

ABMA	American Boiler Manufacturers Association
AISC	American Institution of Steel Construction Manual

ANSI	American National Standards Institute, Inc
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
AWS	American Welding Society
EPA	Environmental Protection Agency
NEMA	National Electrical Manufacturers Association
UBC	Uniform Building Code of USA
IEC	International Electrotechnical commission
NFPA	National Fire Protection Association
IEEE	Institute of Electrical and Electronic Engineers
JIS	Japan Industrial Standard
SPLN	Standard of Perusahaan Umum Listrik Negara

3. DESIGN AND OPERATING CONDITIONS

3.1 General

The HRSG shall be **off fired or unfired**, forced or natural circulation, drum type and vertical HRSG designed in accordance with the applicable requirements of the ASME Boiler and Pressure Vessel Code, Section I- power boiler or other acceptable international standard and shall also comply fully with local codes that may govern.

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The Bidder shall provide all calculations for pressure parts of HRSG which are requested by the applied code and standard.

The HRSG and its associated equipment shall be designed to meet the requirements imposed by the gas turbine and the integrated combined cycle power generation system. The flue gas conditions of the existing gas turbines are at MCR:

Block 2, two GT13E2:		Fuel gas
GT flue gas condition: *)		
Temperature	°C	540
Mass Flow	kg/s	497
Composition mol fraction of:		
N ₂	%	77.55
O ₂	%	12.90
H ₂ O	%	0.75
CO ₂	%	3.8
NO _x	%	0.002
SO ₂	%	0.00
Block 3 each three GT V 94.2:		Fuel gas
GT flue gas condition: *)		
Temperature	°C	539
Mass Flow	kg/s	494.4
Composition % volume of:		
N ₂	%	78.03
O ₂	%	11.90
H ₂ O	%	0.75
CO ₂	%	4.30
NO _x	%	0.01
SO ₂	%	0.000
Block 4, each three GT V 94.2:		Fuel gas
GT flue gas condition: *)		
Temperature	°C	536
Mass Flow	kg/s	494.4
Composition % volume of:		
N ₂	%	75.80
O ₂	%	15.27
H ₂ O	%	0.75
CO ₂	%	3.17
Ar	%	0.001
SO ₂	%	0.000

*) Note : 1. The above data for flue gas condition Block 2, 3 & 4 is a typical data only.

2. The Bidder shall be deemed to have investigated the existing power plant and to have obtained all necessary technical data and information for Bidder's HRSG design purpose, as to risk, contingency and all other circumstances, which may influence or affect to tender.
3. All necessary interfacing works for the proper function of the Combined Cycle Power Plant system shall be deemed to be in the Bidder's scope.
4. The gas turbine compressor before start the performance test
Add on block 2,3 and 4 the filter shall be cleaning by own.

The HRSG shall be capable of following the inherent capabilities of gas turbine to start and shut down rapidly without undue thermal stress.

The HRSG shall be designed to minimize the back-pressure on the gas turbine while maintaining the rated output and steam condition.

The HRSG shall be equipped with a gas bypass system to allow independent operation of the gas turbine when the steam system is not in operation. This facilities will also function to reduce thermal shock in the HRSG while starting without requiring special gas turbine load profiles.

The HRSG shall be designed to accept the gas turbine exhaust gas when fired with gas or oil fuel.

The HRSG shall be designed with gas tight and provided with integral sealed access doors. The access doors shall be so arranged to allow access to the HRSG gas-path, tubing and other pressure parts for minimizing the outage time for inspection and maintenance.

The HRSG shall be designed in a series of large, factory-pressure tested, shippable modules to reduce installation time. Before preassembly of modules at Site it shall be checked. All equipments shall be protected from the corrosion during delivery and erection. Headers and casing shall be welded on Site in a proper manner.

The HRSG units and associated equipment shall be located outdoor. All parts of equipment requiring protection from weather, moisture, and extreme temperature shall be of weather proof construction or shall be provided with watertight shields or enclosures.

The HRSG shall be arranged with the total pressure parts comprising steam drums, superheaters, evaporators, economizer, headers downcomer and integral pipe works in the form of a self-contained unit supported by its own steel structure. This structure is to be independent of any building except for normal points of interconnection with access galleries, platforms or stairways.

The Bidder shall state recommended method based on actual results for safe HRSG start-up and stop operation including initial HRSG and its auxiliaries operation in this Bid.

The Bidder shall also carry out a flow model test or computer modeling of the proposed flue gas system to verify velocity and pressure profiles for vertical HRSG and submit the result to the Owner for approval.

3.2 Design and Performance Data

The HRSG and its associated auxiliary shall be designed for daily Start Stop (DSS) system as Peaking Power Plant.

The repetitive periodic operation shall be required to accommodate the PLN system loading pattern.

The automatic variable pressure operation shall be designed to minimize the steam turbine thermal stress level and obtain the desired flexibility and efficiency.

The Bidder shall define the variable pressure characteristics of the HRSG including the minimum load at which variable pressure operation can be obtained.

Under conditions of steam turbine's trip the steam from the HRSG shall be damped through turbine bypass system to the condenser and transfer to the simple cycle operation mode by means of the exhaust gas bypass by closing the diverter damper.

HRSG and Auxiliary Equipment shall be designed to able to start up in the shortest time in the both case of open cycle circuit mode (the diverter damper is bypass stack open position) and combined cycle circuit mode (the diverter damper is bypass stack close opposition).

The gas side of the HRSG passages shall be designed for the maximum temperature, pressure and mass flow that can be anticipated under all operating conditions including trip conditions.

The selection of pressure and temperature condition for the each pressure steam circuit and the feed water entering the economizer shall be of proven combined cycle plant design to attain the maximum plant heat rate as specified in the relevant section.

The design pressure and temperature for all pressure parts of the HRSG shall take into account the maximum steaming rate and the highest gas temperatures of the gas turbine at its maximum capability condition over the full range of ambient temperatures.

The HRSG shall be designed to be able to start up when the gas turbine is already operated on simple cycle operation. Exhaust gas regulation shall be achieved by regulating the diverter damper.

The Bidder shall design the HRSG and Auxiliary Equipment using good engineering practice and proven experience, in order to obtain good and reliable equipment having superior performance. No prototype equipment or components shall be used.

Further following Design requirements shall be considered.

Performance and Design Criteria	Unit	Data
a. Superheater Materials	-	Superheater Materials shall be designed to be able to start up when the diverter damper is bypass stack close position
b. Approach point at each pressure parts	°C	More than 10°C
c. Pinch point at each pressure parts	°C	Less than 15°C
d. Flue Gas Temperature at stack	-	More than water/acid dewpoint
e. Cold Start-up time to full load (not applicable)	Min (not applicable)	Less than 65 min (not applicable)
f. Warm Start-up time to full load (not applicable)	Min (not applicable)	Less than 50 min (not applicable)
g. Hot Start-up time to full load (not applicable)	Min (not applicable)	Less than 50 min (not applicable)
h. Vibration		- The exhaust duct system shall be designed to avoid drumming, vibration and any resonance with the exhaust gas pulsations under all operating conditions. - HRSG's steel structure and steel supports shall be designed to avoid vibration
i. Noise		Silencers Bypass stack in HRSG to reduce the plant noise level shall provide

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4. DETAILED REQUIREMENTS

4.1 General Design and Constructions

Materials for all parts of the HRSG and its auxiliaries shall be designed, constructed in accordance with the applicable requirements of ASME Boiler and Pressure Vessel Code, Section I- Power Boilers and also comply fully with other government and local codes that may apply.

All walls and tubes shall have wall thicknesses of minimum 10 percent greater than required by Section I of the ASME Boiler and Pressure Vessel Codes together with a suitable allowance where corrosion or erosion can be expected to occur.

All pressure parts of the HRSG unit shall be supported on or suspended from structural steel work and shall be properly braced and stayed to prevent misalignment, sagging, and vibration and to minimize stress.

All necessary access and opposite inspection doors shall be furnished to facilitate removal and replacement of any section of element with minimum interference with any other pressure parts, tubing or fitting not affected by the maintenance.

Double isolation shall be provided for HRSG which may require on line maintenance.

HRSG's steel structure shall meet the specific requirement of Part 4, Technical specifications S-401 and S-402.

4.2 Drums and Headers

The drums shall be of welded construction based on ASME standards or equivalent, equipped with a hinged manhole fitted with an approved steel door at each end.

All drums and headers shall be shop-hydro tested and site-hydro tested.

The drums shall be sized sufficiently large to accommodate drum water level variations during start-up and during operating transient conditions without resorting to wasteful water dumping or risk of carry over, and of sufficient

diameter to ensure easy access for tube cleaning, repairs and inspection of internal fittings.

All nozzles shall be welded to the drum. Necessary connection should be provided on the drums to maintain the drum level along the length of the drum.

The Bidder shall design and locate the point of chemical injection and nitrogen blanketing in such a way as to avoid the by passing of the chemicals to the sampling or blowdown lines.

Adequate drum internal fittings of approved design and manufacture shall be fitted in steam drum and full details with drawings shall be included in the Bid.

Steam purification equipment shall be designed in conjunction with any necessary internal drum driers, deflectors, baffles and distribution pipes in order to reduce the amount of suspended and dissolved solids carried over with the saturated steam to levels acceptable to the steam turbine.

Feed water connections shall be distributed uniformly over the length of the drum and the connections into the drum designed to minimize thermal stress. An approved system of integral feed piping, valves, connections and supports shall be provided to connect the HRSG to the feed pipework.

If necessary anti vortex baffles at the downcomer entrance shall be provided to ensure efficient circulation.

All headers shall be of seamless forged manufacture and shall be provided with arrangements for internal inspection, cleaning and replacement of headers.

Drain connection nozzles shall be welded to the headers or pipings where required.

Common header for each block shall be design connected each other.

4.3 HRSG Flue Gas and Water Flow Paths

The gas turbine exhaust passes through the superheaters, evaporators and economizers all of which are located across the gas stream.

The heating surface of various modules shall be arranged so as to maximize the heat amount received from the exhaust gas with the lowest practical exit gas temperature from HRSG, without risk of damage from corrosive sulphur products at the economizer exit and within the stack.

The feed water temperature at inlet of the economizer shall be selected and controlled for the cases of natural gas and HSD oil firing respectively, to ensure that metal temperature in any part of the economizer shall be always remained above the dew point.

The tubes and headers in each plenum shall be completely drainable and provision shall be provided to gain access to the tubing for inspection and maintenance.

The heat exchanger tube banks of the HRSG shall be mounted in suitable gas tight thermally insulated enclosure, leak-proof to prevent leakage of hot gases and constructed to avoid accumulation of gases and prevent distortion and deterioration of the insulation and steelwork due to high temperature and expansion movements.

Due consideration shall be given to the design of the chamber with the requirements for rapid starting, continuous maximum load and load change. The HRSG enclosure shall be designed to prevent unallowable stresses.

Framing members for the HRSG shall be so arranged that they form a rigid self-containing construction.

The casing shall be designed to avoid drumming or vibration from exhaust gas pulsations under all operating conditions.

HRSG's steel structure and steel supports shall be designed to avoid vibration.

The expansion joints shall be fabric or metal design.

The expansion joints of fabric design shall have fabric bellows elements consisting of an outer layer of Fluoropolymer reinforce with fiberglass, needle or ceramic insulating mats with mesh and vapor barrier. The joint shall be protected internally against flue-gas and externally against rain water.

All expansion joints shall be flanged for removal without disturbing the main sections of the ducting. The Lagging shall be sectionalized for removal local to the expansion joints. Bolts for the flanged joints shall be not less than 15 mm diameter.

Instrument tapping points shall be provided in the casing plates where required for test and monitoring purposes.

4.4 Tubes

The tubes shall be of seamless steel and sized sufficiently large to provide adequate circulation ratio to minimize circulation upsets which may occur during rapid start-up or load change.

Fins added to the heat exchanger tubing to improve the heat transfer characteristics must be continuously welded to the outside surface of the tubes. The fin spacing shall be as required to accommodate the flue gas of natural gas or HSD oil firing.

All welds and tube connections to headers shall be accessible for inspection and maintenance.

All tube welds located within the gas passage shall be hydrostatically tested prior to shipment.

The supporting arrangement shall be designed to prevent any distortion/and any bolts, fittings, spaces or support brackets subject to high temperature gas flow shall be of approved heat and corrosion resisting steel.

Arrangements for expansion of the tubes relative to the unit and their own supports shall be of proven design and provide maximum flexibility to accommodate differential thermal expansion to individual loop within a section.

The elements shall be divided into several banks and the space between banks shall be adequate to permit access for maintenance.

4.5 Superheaters

The superheater shall be designed such that the steam temperature at the steam turbine inlet will not exceed the allowable steam turbine parts limit temperature over any load.

The design shall be compatible with the requirements of variable pressure operation and the variable characteristic of the turbine exhaust gas flow at all loads. The arrangement of superheater heating surface material shall be in accordance to ASME and ASTM standards.

Provision shall be made in all headers and interstage piping to accommodate test pressure gauges and quick response thermocouples. The HRSG wall shall have seal probe pockets to allow insertion of portable thermocouples for gas temperature measurement at each superheater inlet.

An approved number of chromel-alumel thermocouple shall be supplied for measuring and recording metal temperature during pre-commissioning tests and normal operation on a temperature recorder provided under this Contract.

All superheaters shall be shop-tested and site-tested.

The Bidder shall state the method of controlling steam temperature at the superheater.

The Bidder shall state the anticipated variation in the superheater steam temperature and tube metal temperature possible in relation to design temperature.

4.6 Evaporators

The evaporator plenums shall be designed such that the steam generation rate could be achieved with a maximum pinch point of not more than 15° C, which shall be stated in the Bid.

The pinch point is defined as the temperature difference between the flue gas temperature leaving the evaporator zone and the saturation temperature of the steam at the evaporator.

If the forced circulation applied for the evaporators, each evaporator shall be equipped with the 2 x 100% capacity circulation pumps which are installed between the drum downcomer and the evaporator feed inlet headers.

The evaporator shall be designed to operate over the full load range of the HRSG and ensure an even distribution of water through the tubes.

All evaporators shall be shop-tested and site-tested.

4.7 Economizers

The economizers shall be designed to ensure there will be non-steaming feed water flow operation through the full operating range of the HRSG alternatively Bidder shall provide a system with feed control station located downstream of economizers and provide bypass especially in oil firing mode to avoid the steaming in the LP economizer.

Feed water from the deaerator storage tank to the economizer of HRSG shall be supplied by the boiler feedwater pumps.

Provision shall also be made to prevent the tubes in either economizer from siphoning dry.

All economizers shall be shop-tested and site-tested.

4.8 Blowdowns and Drains

The steam drum shall be provided with a continuous HRSG water blowdown connection, complete with parallel slide isolating and regulating valves in accessible positions adjacent to the drum connection, capable of controlling the rate from 0.05% minimum to 5% maximum of the HRSG steam rating. Where necessary, all drainable sections of the HRSG shall be provided with intermittent blowdown and drain piping.

A blowdown tank shall be sized for the maximum continuous/intermittent blowdown flowrate.. All drains requiring traps shall be fitted with necessary isolating valves located in accessible positions. No blowdown or drain pipe shall less than 40 mm bore.

The blowdown tank shall be equipped with appropriate silencer at the blow pipe to atmosphere.

Where atmospheric waste drains are fitted they shall be grouped as far as practicable and led to approved locations at ground level.

The Bidder shall submit calculations for sizing of the blowdown tank and vent pipes.

4.9 Valves

4.9.1 Safety Valves

The Bidder shall comply with the requirements of M-004 valves.

Safety valves of shall be provided in accordance with ASME codes or equivalent requirements and this specification. The safety valves shall be arranged to blow off in a predetermined sequence.

The superheater safety valves to blow firstly at the superheater outlet shall be of the automatic electrically actuated type.

Providing that the turbine by pass systems have a full relief duty of steam flow from HRSG under all operational and fault conditions, no electrically actuated safety valve will accepted.

Double isolation valves shall be provided to allow maintenance to be carried put on the electrically operated safety valves when the HRSG is in service.

The safety valves at the super heater outlet shall be sized to have a discharge capacity equal to at least 20% of the maximum steaming rate of the HRSG.

The remaining valves are collectively to have the maximum total discharge capacity required for the protection of the HRSG.

An approved system of safety valve escape piping, drains, supports and silencers with expansion joints, anchors and supports shall be provided.

A separate escape pipe and discharging to atmosphere shall be provided for each safety valve. All such piping shall be terminated to prevent a personnel hazard.

For each enclosed safety valve chest/an unrestricted means of draining shall be provided. The drain pipe shall be led to a tundish fitted with a hinged cover. Tundishes shall be located at a convenient level local to gallery and the outlet pipe from the tundishes shall be led into a common pipe and routed to the drain pipe work serving the blowdown tank.

All safety valve escape pipes shall be fitted with independent silencers. The silencers shall be adequately designed to resist the vibrational stresses set up by the escaping steam. Where they pass through the roof sheeting, suitable upstands should be provided in the sheeting to prevent rainwater ingress into the building. Suitable weather hoods on the pipes shall also be provided.

4.9.2 Blowdown Valves

An adequate number of blowdown valves and superheater drain valves shall be provided.

Each blowdown and drain point shall have two steel valves in series of approved size and diameter.

The upstream valves shall be of manual operated type, and second valves shall be electrically operated type.

For the blow off valves where the pressure and flow are always in one direction, the first isolating valve in each case shall be parallel slide type.

4.9.3 Drain Valves

The necessary drain valves of an approved type and size shall be provided for draining the HRSG completely within one hour. A free blowing drain shall be provided between the main stop and non-return valves.

4.9.4 Main Steam Stop Valves

The main steam stop valves shall located as close as practicable to the HRSG and shall be provided with motor and hand actuation complete with pressure equalizing bypass valves at each main steam outlet from the superheater.

4.9.5 Main Steam Non-Return Valves

Main steam non-return valves of anti-chattering swing type check valves shall be provided at each main steam stop valve outlet, upstream of the main steam stop valve outlet.

4.9.6 Feed Check Valves

Anti-chattering check valves shall be provided at each economizer inlet.

4.9.7 Air Release Valves

Motor operated air release valves are to be provided where necessary, with safety discharge. A U-pipe is to be fitted on each valve outlet.

4.10 Connection for Fuel Preheater

Pipe, valve, flange, gasket and other fittings shall be prepare at each HRSG unit to stream the steam/hot water which will be used for existing gas turbine fuel preheater and the steam/hot water shall be returned back to HRSG.

4.11 Steam and Water Sampling

An adequate number and type of connections with valves, fittings and internal piping shall be provided on each steam drum and the economizer inlet, so that representative samples of HRSG water can be taken, utilizing a sampling assembly.

An adequate number of nozzles shall be provided and fitted in approved positions for the purpose of sampling drum steam.

The sampling points shall include steam at the outlets of the superheaters.

Each sampling point shall be complete with a nozzle and all necessary valves and connections.

Nozzles shall be of iso-kinetic type, designed and located so that representative samples can be taken from the approved point.

4.12 Chemical Dosing

An adequate number of chemical injection points shall be provided at the steam drums.

The connections shall be designed to minimize thermal stress. Connections into the drum shall be complete with a distributor inside the drum shell to ensure preferential distribution into the incoming feed water.

4.13 HRSG Mountings, Fittings and Integral Piping

The HRSG shall be provided with the mountings and fittings of approved standard to ensure safe and efficient operation and to meet the requirements of the specified standard and regulation. They shall be of cast or forged steel and suitable in every respect for the temperature and pressure conditions under which they will operate.

The mountings and fittings shall be located in approved positions and fitted with name plates as specified elsewhere.

All mountings, fittings and integral and other piping connected to the HRSG shall be adequately supported to avoid transmission of vibration stress to the welded connections on the steam drums or headers.

To facilitate on-load maintenance, double isolation shall be provided in connection for the application of its operating pressure is 50 bar gauge or above.

4.14 HRSG Framing and Support

If the HRSG with its integral parts is supported by a steel frame work it shall be of ample strength, stiffened and braced. The structure shall be provided with steel bases properly secured to the foundations.

The Bidder shall state in his Bid the location and magnitude of all loads imposed on the foundation by the supports.

The HRSG shall be supported independently so that it is free to expand and contract without adverse affect.

Where sliding supports are used they shall be lubricated from accessible positions.

Steel work in contact with hot air or gases or subject to radiation from any part of the HRSG shall be insulated.

The Bidder shall state in his Bid for obtaining a weather proof arrangement. Any such design shall allow full freedom of expansion of the HRSG.

4.15 HRSG Insulation and Cladding

The HRSG shall be insulated and clad externally to provide fully weather proof units suitable for outdoor operation.

The insulation shall be of approved material and suitable for continuous service at the maximum operating temperature.

All external insulation shall be in accordance with Part 4, Specification M-003.

The material used shall be suitable for the adverse climate condition prevailing at site and for the proximity of the site to the sea.

Where internal insulation is applied, the Bidder shall described the detail of insulation system to be furnished. Internal insulation liner shall be stainless steel.

4.16 Galleries, Staircases, Platforms and Ladders

Easy access to all valves, gauges, instrument fittings, observation points, access doors and for all other operational purposes shall be provided by stairs, platforms or walkways adequately safeguarded with hand-railing and toe-plates.

Ladders shall only be used for creating extra means of escape.

A minimum of one set of stairways and galleries shall be provided for HRSG. They shall extend from the basement level to the highest parts of the units. Galleries shall also be provided along the front and rear of HRSG drums.

Uninsulated metal roof shall be provided to cover the each pressure drum galleries.

The Bidder shall be also provide ladders in addition to the one stairway. Ladders to shall be provided on opposite side from the stairway.

4.17 HRSG Stack

The exhaust gases from HRSG shall discharge through self-supporting steel Stack.

It shall be of approved design and constructed according to the approved standard with sufficient height to ensure that the ground level pollutants are within the limits specified in Clause 2.16 Environmental Considerations in section 2 of part 4 and also 'Peraturan Menteri Negara Lingkungan Hidup No.21, 2008 tentang Baku Mutu Emisi'

A dispersion study to determine the stack height shall be performed by the Bidder in order to meet the ambient air standard as mentioned in Part 4, Subsection 2.16, however the minimum height shall be 45 m above ground level.

The Stack shall be clad over their full height with an aluminium alloy of approved grade and thickness. It shall be insulated entirely to ensure the internal metal temperature of the stack above water/acid dew point on all load. In addition, where personnel may access, personnel protection insulation shall be provided. The cladding shall be painted with red and white bands to meet local aircraft safety requirements

The Bidder shall provide a motor operated damper to prevent entry of rainfall into the HRSG under maintenance or to keep optimum temperature during HRSG shutdown.

The Bidder shall state clearly his proposal for achieving the blanking of the HRSG stack which may include,

- An electric actuator drives the damper blades, with a drain, located in between HRSG and stack.
- The drives shall be designed such that in the event of over pressure on the flue gas side, the blades are free to open.
- The Bidder shall state how the blade is assembled, located and removed during the maintenance period.

The Platform shall be designed and installed at the top of stack for installation and maintenance.

The stack shall be equipped with obstruction lighting, lighting protection and earthing.

The studies/calculations/analysis for wind oscillation effect on stack shall be performed by the Bidder

4.18 Painting

The painting for the HRSG and Auxiliary Equipment shall meet the specific requirement of part 4, Technical Specifications M-001, S-420 and S-430.

4.19 Internal Chemical Cleaning of Pressure Parts

After satisfactory completion of the general cleaning, the complete HRSG unit shall be flushed and boiled out using water in which an approved alkali chemical has been added. The process shall be carried out for a period which will ensure a clean internal surface, then the entire unit shall be drained and washed out with demineralized water.

The waste water in chemical cleaning stages shall be neutralized and taken away by the Bidder.

Chemical cleaning procedures for the HRSG shall be submitted to the Owner for approval.

Experienced site personal shall be provided, to supervise all chemical cleaning and verify the adequate chemical cleaning is accomplished.

Following HRSG chemical cleaning nitrogen gas shall be used to purge all air from the pressure parts.

Suitable blanked off connections shall be provided on the HRSG to enable post service acid cleaning to be undertaken when required.

After internal chemical cleaning, steam blow shall be performed in accordance with Part 4, Subsection 4.2 "Mechanical Work".

4.20 Nitrogen

A complete nitrogen gas piping system including sufficient number of nitrogen cylinders shall be furnished to preserve HRSG and other equipment in a dry condition.

The Bidder shall provide an isolating valve adjacent to each item of equipment connected to nitrogen gas header.

The Bidder shall provide flexible hoses at each connection.

4.21 HRSG Storage

The Bidder shall be responsible for ensuring the HRSG is maintained in a passive state until it can be put into commercial operation. The Bidder shall provide chemicals, etc., necessary to maintain the HRSG in a non-corrosive state in accordance with the clause on boiler cleaning.

Demineralized water and nitrogen for storage of the HRSG shall be provided by the Bidder

4.22 Gas Fuel Preheater (Not applicable)

The Bidder shall be provide and install the facility gas fuel preheater in each HRSG.

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5. PERFORMANCE AND RELIABILITY TESTING

5.1 General

Performance and reliability testing shall be in accordance with the requirements of Part 4, Subsection 2.6 Performance Testing and Guarantees and as specified herein and Part 2, Clause 1.

5.2 Performance Test

5.2.1 Objective and Test Method

After the HRSG unit and its auxiliary equipment are placed in service, the Bidder shall conduct the performance test of the HRSG in the form of combined cycle plant. The HRSG performance shall be verified in accordance with the latest ASME Power Test Code 4.4 for the owner's reference.

5.2.2 Condition of HRSG

The tests shall be made with HRSG heating surfaces in new and clean condition and the tests shall be made not later than four weeks after HRSG's gas passages have been internally cleaned and inspected. Any additional cleaning between time of internal inspection and test shall be made with cleaning equipment provided by the Bidder.

5.2.3 Scheduling of Test

The Bidder shall establish the schedule and detail of the testing as per Part 4 Technical Requirements, Subsection 4.1 General Requirements. The duration of the Reliability Run of the Plant shall be as per Part 2, Clause 1.

5.2.4 Performance Parameters

The following items, but not be limited, shall be confirmed in the performance test:

- HRSG each pressure circuit capacity based on feedwater flow
- HRSG efficiency given in part 5 shall be computed in accordance with ASME PTC 4.4. the heat loss method will be used for the test.
- Gas resistance through complete unit
- Steam pressure at HRSG outlet for each pressure circuit
- Pressure drop between HRSG terminal points (feedwater inlet to main steam outlet)
- Superheater outlet temperature control range
- Steam purity for each pressure steam circuit

5.3 Deficiencies

The Bidder shall remedy any deficiencies observed during the performance test at his expense. This shall include but not be limited to the addition, deletion or rearrangement of any heat transfer surfaces, structural modifications, rerouting of external piping, rearrangement of galleries, rerouting of electric cable trays or conduits, modification of control systems as necessary to meet the technical requirements.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality control shall conform to the requirements in Part 4, Subsection 2.7.

The quality control procedures shall include, but not be limited to, the following:

- Welding
- Non-destructive testing
- Heat treating
- Packaging and shipping
- Shop Test Procedures
- Cleaning and Painting

The quality control documentation shall include, but not be limited to the following:

- Shop Test Reports
- Material Certifications
- Non Destructive Examination (NDE) Reports
- All other documentation required by applicable codes and standards

Bidder's quality control activities shall generally conform to the following requirements, as the minimum:

Table 6.1.1-(1) Quality Control Activities

(0: Applied)

Item	UT: Ultrasonic Testing	MT : Magnetic Particle Testing	Quality Control Stages and Activities	Documentation and Records
Steam Drum				
(a) Steam Drum	O (for flat plates)	O (for weld seams)	• Inspection flat of plates • Hot form half shells • Fit halt shells together • Weld longitudinal seams to make shell • Fit shells together • Weld shell-to-shell circle seams • Weld downcomers to shell	
(b) Steam Drum Heads	O (for flat plates)	-	• Inspection of flat plates • Hot form hemispherical heads • Inspect forming • Fin and weld nozzles and downcomer transition piece	

Table 6.1.1-(1) Quality Control Activities

(0: Applied)

Item	UT: Ultrasonic Testing	MT : Magnetic Particle Testing	Quality Control Stages and Activities	Documentation and Records
(c) Steam Drum Assembly	O (for stub welds)	O (for weld surface)	• Fit heads to drum • Weld heads • Hydraulic test and final inspection • Presentation for shipment	
(d) Verification records	-	-		• Material certifications • Welding procedure and qualification of procedures and welders and weld operator's certifications • Heat treat furnace charts • Calibration record • NDE personnel certifications • All radiographs, reader sheets and shooting sketches • Cleaning

Table 6.1.1-(1) Quality Control Activities

(0: Applied)

Item	UT: Ultrasonic Testing	MT : Magnetic Particle Testing	Quality Control Stages and Activities	Documentation and Records
Header down comers, piping				
(a) Headers, down comers, major piping	-	-	weld girth seams	
(b) verification records	-	-		- Material certifications - Welding procedures and qualification of procedures welders and welding operator's certifications - All radiographs, reader sheets and shooting sketches - Heat treatment furnace charts - Cleaning

Table 6.1.1-(1) Quality Control Activities

(0: Applied)

Item	UT: Ultrasonic Testing	MT : Magnetic Particle Testing	Quality Control Stages and Activities	Documentation and Records
Casing and tube sheets				
(a) Casing and tube sheets	-	O (for weld joining attachment to panel and tube-to-headers wells)	• Butt weld tube to required panel lengths • Weld tubes-to-headers	
(b) Verification records	-	-		• Material certifications • Welding procedures and qualification of procedures and welder and welding operator's certifications • Heat treat furnace charts • All radiographs, readers sheets and shooting sketches • NDE personnel certification • Cleaning and painting

Table 6.1.1-(1) Quality Control Activities

(0: Applied)

Item	UT: Ultrasonic Testing	MT : Magnetic Particle Testing	Quality Control Stages and Activities	Documentation and Records
Riser and Supply Tubes	-	O (for tube bends)		
Superheater and Economizer section and platena				
a. Superheater and Economizer section and platena	-	-	• Final inspection	
b. Verification records	-	-		• Metal certifications • Welding procedures and qualification of procedures and welders and welding operator's certifications • Heat treatment furnace charts • All radiographs, reader sheets and shooting sketches

Table 6.1.1-(1) Quality Control Activities

(0: Applied)

Item	UT: Ultrasonic Testing	MT : Magnetic Particle Testing	Quality Control Stages and Activities	Documentation and Records
Heat recovery steam generator assembly	-	-	After complete of erection, the heat recovery steam generating unit shall be hydrostatically tested by the Bidder in accordance with the applicable code	
Structural Steel				
(a) Structural Steel Fabricated from Structural Shapes	-	-	• Final fit-up • In-process inspection of painting and shotblasting	
(b) Fabricated Plate Girders			• Final assembly inspection • Inspect painting	
Back stays			• Final inspection	
Flues, ducts, casing (inspect)			• Final inspection for dimensions, welding, workmanship	
Damper (inspect)			• Final inspection	

Table 6.1.1-(1) Quality Control Activities

(0: Applied)

Item	UT: Ultrasonic Testing	MT : Magnetic Particle Testing	Quality Control Stages and Activities	Documentation and Records
Expansion joints			• Soap test welds for leaks • Final inspection for dimensions, welding, connection plates, skirts, pressure stay, painting and workmanship	
Feed and check, stop, safety vent and drain valves (witness or inspect)				
(a) Hydrostatic test, seat leakage test and dimension check			• Hydrostatic test, seat leakage test and dimension check	
(b) Verification records				• Material certifications • All radiographs, reader sheets and shooting sketches and applicable • Cleaning

Additional audit shall be performed at heat recovery steam generator manufacturer's facility:

- Audit receiving inspection, documentation and un priced copies of purchase order to their vendors
- Audit final inspection and documentation
- Audit the manufacturer for the items listed in Owner Quality Assurance Articles to very conformance

Documentation : Three (3) copies of the following documents shall be submitted to the owner or his authorized inspection representative for review and acceptance:

- Material certifications submitted prior to start of fabrication
- ASME code reports
- Test reports
- Non-destructive examination reports
- Certification of non-destructive examinations personnel
- Radiograph : two (2) sets of radiograph film, reader sheets and shooting sketches for each location shall be submitted to the Owner for review and acceptance. Film will be returned to the Bidder for storage
- Copies of all deviations and accepted deviation reports

7. INSTRUMENTATION AND CONTROLS

7.1 General

The Bidder shall be responsible for the complete design, supply and installation of all instrumentation and control systems and devices listed and described in this Technical Specification and Part 4, Technical Specification I-001.

The Bidder shall furnish all control systems for the operation of the equipment being supplied as called for in this Technical Specification. Control systems shall include the latest hardware and state-of-the-art control functions that provide for the best control of the equipment to perform its intended functions.

The Bidder shall furnish all in-line devices and accessories to prevent water damage to steam turbine used for Electric Power Generation including but not limited to desuperheater spray water block valves and steam piping drain pots and drain valves.

The control, monitoring and alarming of equipment shall be implemented within DCS. Field input devices, final control devices and control strategies shall be incorporated into interlock, alarm, protect and control the HRSG.

The Bidder shall provide all instrumentation and control devices necessary to enable the DCS to protect, control monitor and alarm the HRSG equipment and auxiliaries in the control room.

Those control and instruments devices requiring field installation shall be provided with the necessary wiring, piping, tubing and mounting details drawings and instructions necessary to effect installation, electrical connection and calibration.

7.2 Technical Requirements

Technical requirements, governing the design and supply of instrumentation and control systems and devices shall be as detailed in Part 4, Technical Specification I-001, I-002 and I-100.

7.3 Local Mounted Instrument Panels

Local mounted instrument panels shall be in accordance with the requirements of Part 4, Technical Specification I-003.

7.4 Local Control Panels

Local control panels shall be provided as required for testing and maintenance purposes complete with all operator interfaces and shall be in accordance with the requirements of Part 4, Technical Specification I-002 and E-200.

7.5 Thermocouples

The Bidder shall furnish sufficient permanent thermocouples for measuring the temperatures of gases to ensure that the HRSG is protected against overheating and for temperature monitoring during start-up and load operation.

The Bidder shall provide temperature elements to measure HRSG gas temperatures between all heat transfer modules.

The Bidder shall furnish necessary chromel-constantan thermocouples for the steam drum for measuring in the range of from 20°C to the maximum expected temperature to ensure that the HRSG parts are protected against overheating and for steam temperature monitoring during startup and low-load operation.

7.6 Vibration Monitoring

The Bidder shall furnish and install vibration measuring equipment for feed water pumps in accordance with the requirements detailed in Part 4, Technical Specification I-001.

7.7 Instrumentation

The Bidder shall provide instrumentation, devices and equipment to provide complete control and monitoring system for HRSG

- “Smart” type electronic transmitters, including calibration devices
- Pressure and differential pressure switches
- Pressure and differential pressure gauges
- Flow measuring elements
- Thermocouples, resistance temperature detector (RTD) and thermowells

Note : Thermocouples used in averaging, differential or any temperature measuring scheme, in which two or more elements are used in an electrical circuit, shall be of the ungrounded design.

- Temperature switches if required by design, shall utilize thermocouples or RTD in accordance with specification as inputs to locally mounted digital indication switches.
- Temperature gauges
- As a minimum, the following instruments shall be provided by the Bidder for each drum.

Drum water level gauge : One per steam drum

Drum water level transmitter : Each pressure steam drum (three, two, (for control and protection) for one end, one for opposite end)

Pressure transmitter : Each pressure steam drum (one for steam side)

Pressure gauges : Each pressure steam drum (one for steam side)

- Feed water/steam path thermocouples located on the inlet and outlet headers
- Two thermocouples shall be provided at each desuperheater outlet for input the DCS for spray valve control and alarm Gas path pressure test points, with valved connections, shall be provided at the HRSG inlet, outlet, and HRSG access lane.
- On line steam leakage detector HRSG
- Control valves
- Motor operators
- Drum level indicators, for the purpose of local drum level indication shall be provided, including local electronic unit with integral display.

7.8 Installation Requirement of Instrumentation

Instrumentation shall be provided and installed in accordance with standard installation details. All instrument installation details shall be subject to Owner's approval. Instrument tubing and fittings, instrument valves and manifolds specified on installation details shall be provided by the Bidder.

Instrumentation on Bidder's supplied HRSG's and auxiliaries shall be shop assembled to the maximum extent practical. Instrument support structures, such as gauge panels, shall be attached directly to the skid steel or structure. In no case shall the Bidder attach an instrument or instrument support to handrails, enclosure walls or any other non-primary steel member.

Instrument tubing shall conform to Part 4, Subsection 4.5.

Instruments attached to Bidder equipment shall not interfere with equipment access or removal areas nor be located in aisle ways. The Bidder shall provide sufficient clearance area for site personnel to perform maintenance, calibration or testing with instrument in place. The installation design shall not require removal of access housing or covers to disconnect and remove an instrument from service, nor shall the design impede site personnel from easily viewing any instrument provided with local indication.

All instrument valves shall be installed and oriented so that a minimum clearance of 5 cm exists between the handle turning radius and any physical obstruction including the turning radius of any adjacent valve handles.

All pressure instruments shall be provided with facilities for testing, calibration and bleed-off in their installed positions. These facilities may be configured from fittings, unions, valves, etc., or from instrument gauge valves with these integral facilities (valve manifolds). Instruments on the equipment may be mounted at the process taps provided they are accessible and maintainable from the nearest permanent platform and are not subject to vibration or process pulsations, such as at the discharge of a pump.

Redundant pressure and level elements required for control (such as steam drum) shall utilize separate process connections.

The Bidder shall provide connections for all instrumentation and devices necessary for performance monitoring of the HRSG in accordance with ASME Power Test Code PTC 4.4 utilizing the input/output method.

The Bidder's instruments and device accuracies, and error limits shall conform to the PTC 4.4 requirements and/or this specification's requirements, whichever are most stringent.

The level gauge at each end of the HRSG drum shall be fitted with double isolation valves and fixed so that the water level can be easily seen.

The gauges shall be fitted with separate drains and self-closing check valves for shutting and blowing. The gauge glass fittings shall be of forged steel and shall be so designed that they can be removed as a complete assembly.

The connections to the water level gauges and indicator shall be arranged so that any gauge can be isolated for maintenance without affecting the working of the other gauges.

Signal for high and low level alarm or trip be produced in the DCS using signal from three drum level transmitters.

The two out of three logic shall be applied to low-low drum level for trip of the HRSG and damper control operation, and high-high drum level for trip steam turbine.

7.9 Control Requirements

The Bidder shall provide (for instrumentation and control devices and final control elements) calculations, algorithms, operational descriptions, etc., necessary for control and monitoring at DCS.

All the DCS analog interfaces between local and DCS shall utilize 4-20 m ADC signals.

Diaphragm actuated control valves shall be provided in accordance with Specification M-005, including air sets, positioners, pilots solenoid valves, limit switches, I/P converters, and position transmitter, for operation by DCS. Controlling functions shall be implemented in DCS.

Air operated piston block valves shall be provided with solenoid valves and limit switches for operation by DCS.

The Bidder shall provide any isolation or surge protection devices as necessary to protect HRSG systems and devices from the high lightning activity associated with the Muara Tawar site area. Additional requirements suggested or added by the Bidder for HRSG's lightning protection or grounding system shall be provided.

The Bidder's design shall incorporate two grounding schemes; an equipment ground for all cabinets, enclosures, steel, etc., and an instrument ground for device shield grounding, low level signal grounding, etc. All shields shall be terminated to the DCS instruments ground by Bidder. No instrument device shields shall be grounded at Bidder's equipment or instruments.

7.10 Flue Gas Monitoring

The Bidder shall supply all gas monitoring connections for continuous emission monitoring (CEM) equipment on the stacks.

Continuously monitoring shall be carried out for the entire operational life span of the plant. The operating of a gas-fired power plant shall be supported

by a continuously working online monitoring system for measurement of nitrogen oxides, carbon monoxide, carbon dioxide and oxygen.

In case of oil-firing a monitoring system for sulfur dioxides and particulates shall be added.

The particulates matter in the flue gas and the smoke density (opacity) shall be measured by hand analysis based on the Decree of Head of Environmental Management Agency No.kep.205/Bapedal/07/1996 concerning Technical guidance for air pollution control, Bapedal 770-0005-1996 and Bapedal 770-0006-1996 respectively.

In addition to these relevant parameters, the online monitoring equipment shall also be able to measure the temperature, velocity of the flue gas which will be released to the stack.

In-situ sampling type analyzers could be used for monitoring nitrogen oxide (NO_x) carbon monoxide (CO), carbon dioxide (CO₂) sulfur dioxide (SO₂). These analyzers shall be provided with on-line calibration capability.

Non-Dispersive Infrared (NDIR) type gas analyzers for NO_x, CO, CO₂, SO₂ and Magnetic wind methods type gas analyzer for O₂ shall be selected.

Permanent access shall be provided to stack for above exhaust gas monitoring and the particulates matter. The sampling ports shall be located at least eight stack diameters downstream of any bends, inlet, constrictions, abatement equipment, or other flow disturbances and at least two stack diameters upstream of the stack exit or other flow disturbances as stipulated by the Decree of Head of Environmental Management Agency No.kep.205/Bapedal/07/1996 concerning Technical guidance for air pollution control, Appendix I.

The exhaust gas sampling shall be performed at HRSG stack and bypass stack.

The change-over of the exhaust gas sampling for HRSG stack and Bypass stack shall be performed by remote control operation.

Flanged connection and connection shall be provided in accordance with U.S EPA requirements.

Access platform shall be provided as well as insulation at the monitoring connection for personnel protection.

8. ELECTRICAL SPECIFICATIONS

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the HRSG and auxiliaries as listed in Clause 1 of this Technical Specification M-301 HRSG and auxiliary equipments.

The electrical equipment and/or accessories shall include but not be limited to, the following :

- Motors
- Local control boxes
- Conduit, cables and wiring
- All electrical control, monitoring and instrumentation devices and other appurtenance.

All electrical equipment located in the HRSG area shall be suitable for outdoor operation.

For general electrical requirements, see Part 4, Technical Specification E-001.

8.2 Motors

The motors shall meet the following specific requirements as well as the applicable general requirements of Part 4, Technical Specification E-210 for Low Voltage Motors.

8.3 Conduit, Cables and Wiring

The cable furnished by the Bidder shall meet the specific requirement of Part 4, Technical Specifications E-230, E-231 and E-232.

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